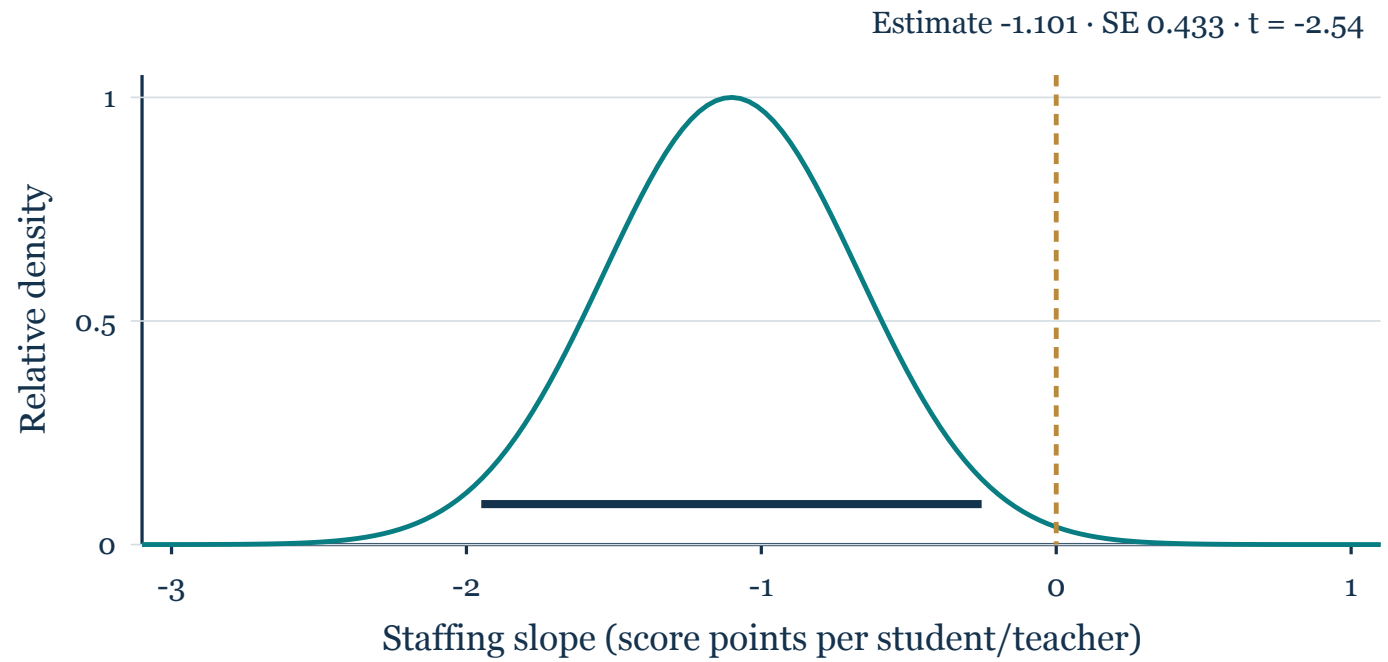
Is the staffing relationship large enough to matter?

A decision needs magnitude and uncertainty.



Actual staffing estimate and robust SE. The reference curve is a large-sample normal approximation, not the distribution of district outcomes.

> Question: how accurately does the sample measure the relevant magnitude?

> Inference adds an uncertainty argument to the fitted estimate.

Recall which coefficient the model estimates

Keep its units and adjustment fixed.

> The adjusted staffing slope is -1.101 points per additional student per teacher.

> Before calculating a test, state the exact target.

An outcome distribution is not a sampling distribution

The random object changes.

> A distribution describes the possible values of a random quantity and how often they occur.

> These are different objects.

Describe spread before estimating uncertainty

Variance and standard deviation summarize different kinds of variation.

> $\mathrm{Var}(0, 1, 2) = (1 + 0 + 1)/3 = 2/3$

> Standard deviation: $\sqrt{2/3} \approx 0.816$

> Residual degrees of freedom: $n - k$

A standard error estimates the estimator’s spread

It is not the standard deviation of the outcome.

> SE: estimated repeated-sample standard deviation of the coefficient.

> Normal reference: about 95% between -1.96 and 1.96.

State a null before calculating a p-value

The evidence is evaluated under a specific claim.

> $H_0 : \beta_1 = 0, H_1 : \beta_1 \neq 0$

> $t = (\widehat{\beta}_1 - \beta_{1,0}) / SE(\widehat{\beta}_1)$

> p-value: a tail probability under the null and reference assumptions.

Build a confidence interval step by step

Combine the estimate with a stated reference rule.

> $\widehat{\beta} \pm 1.96\,\mathrm{SE}(\widehat{\beta})$

Confidence interval

Try the comparison before opening the explanation.

Practice

An estimate is 0.30 with robust SE 0.12. Use the large-sample 1.96 rule to form a 95% interval and test a zero effect.

Confidence interval: explanation

Follow the calculation and interpret its units.

> $0.30 \pm 1.96 \times 0.12$ gives [0.0648,0.5352]. The t ratio is 2.5, so reject zero at approximately 5% under the relevant large-sample assumptions. This says nothing by itself about the economic importance or causal validity of the estimate.

A rejection is not an effect-size statement

Statistical significance and practical value answer different questions.

> A statistically significant result passes a chosen testing threshold.

> For a staffing decision, translate the slope into a feasible ratio change and compare the implied learning difference with costs and relevant alternatives.

Normal-error exact inference and robust approximations differ

Know which argument supports the reference distribution.

> Exact conventional t/F: normal, independent, equal-variance errors under the linear model.

> Robust: large-sample approximation allowing unequal variance; dependence needs separate treatment.

Unequal error spread is heteroskedasticity

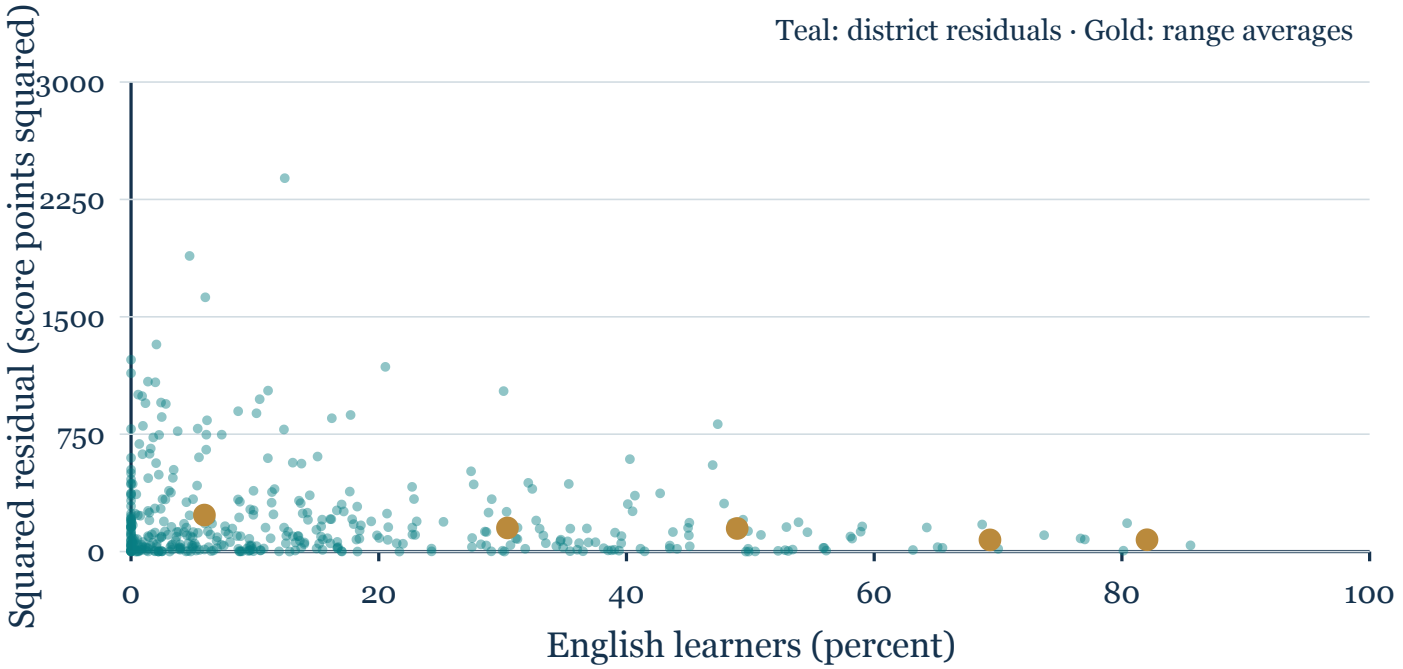
It concerns the error variance at different regressor values.

> $E(u_i \mid X_i) = 0$ concerns the mean

> $\mathrm{Var}(u_i \mid X_i)$ concerns spread

Look for a variance pattern before choosing a remedy

A squared residual is an observable clue about error spread.



Actual OLS residuals from 420 districts. Gold points average squared residuals in five language-share ranges; these are diagnostic summaries, not a known variance curve.

- Residual: observed outcome minus fitted outcome.
- Squared residual: $\widehat{u}_i^2$.
- Gold: average squared residual within each of five language-share ranges.

Breusch–Pagan turns the variance question into a test

The null is constant conditional error variance, given the chosen mean model.

```
. estat hettest str el, iid

Breusch–Pagan/Cook–Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: str el

H0: Constant variance

    chi2(2) =  29.50
Prob > chi2 = 0.0000
```

Actual StataMP output; reproduce with variance.do in the Stata download.

- $H_0 : \mathrm{Var}(u_i \mid X_i) = \sigma^2$: constant variance; X_i denotes the included regressors.
- Regress $\widehat{u}_i^2$ on a constant, staffing and language share.
- Test statistic: nR^2_{aux}, compared with χ^2_2.

White allows a broader variance pattern

Squares and products expand the auxiliary variance regression.

```
. estat imtest, white

White's test
H0: Homoskedasticity
Ha: Unrestricted heteroskedasticity

    chi2(5) =  42.46
Prob > chi2 = 0.0000
```

Actual White-test block. The complete diagnostic output is in variance_output.txt.

- Auxiliary columns: staffing, language share, their squares and their product.
- White statistic: nR^2_{aux}, compared with χ^2_5.
- Distinguish a variance diagnostic from a test of the causal comparison.

Practice

Why do BP and White have different degrees of freedom here?

Explanation: BP uses two auxiliary slopes; White adds two squares and one product, giving five.

Weighted least squares changes how the line is fitted

Give less weight to observations with greater conditional error variance.

> WLS minimises $\sum_i w_i (y_i - a - b_1 STR_i - b_2 EL_i)^2$.

> Candidate model: $\mathrm{Var}(u_i \mid X_i, n_i) = \sigma^2 / n_i \Rightarrow w_i = n_i$.

> Multiply y_i, every regressor and the constant by $\sqrt{n_i}$.

Read a weighted fit without assuming it must be better

A variance model and an economic target both matter.

```
. regress score str el [aw=enrol], vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |   -1.32611   .6605944    -2.01   0.045     -2.62462   -.0276004
          el |  -.6390283   .0408085   -15.66   0.000    -.7192442   -.5588124
       _cons |   693.1913   13.26319    52.26   0.000     667.1203    719.2624
------------------------------------------------------------------------------
```

Actual StataMP output; reproduce with variance.do in the Stata download. Coefficient table shown; the download retains the full model output.

- OLS and WLS use the same 420 districts.
- The transformed regression reproduces the weighted coefficient.
- Weighting does not remove an omitted common cause.

Practice

Which column is easy to forget when transforming the weighted model?

Explanation: Multiply the constant column by the square root of enrolment as well.

Generalised least squares includes variance and dependence

WLS is one special case of a broader transformation idea.

> GLS: transform using the assumed error variances and covariances.

> Feasible GLS: estimate that error structure before transforming.

> Choose separately: mean specification, estimator and uncertainty calculation.

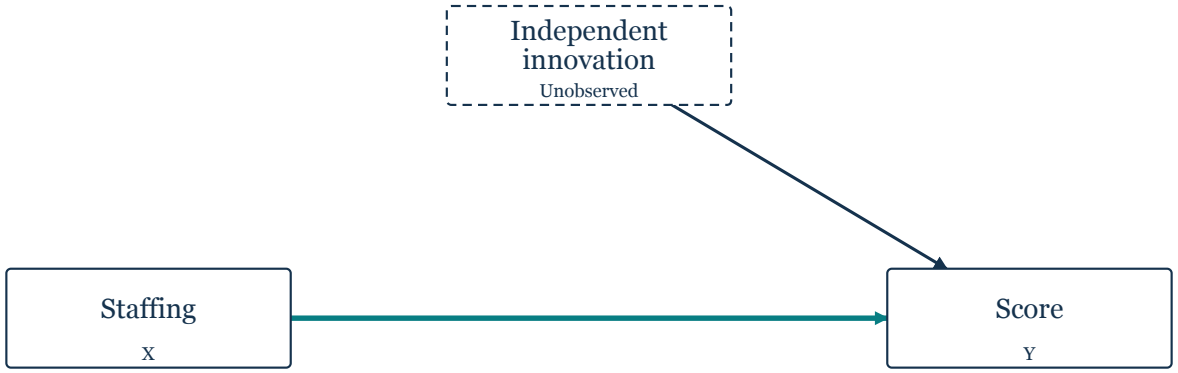
Practice

Would weighting remove an omitted cause related to staffing and scores?

Explanation: No. Justify the causal comparison separately from the error-variance model.

A DAG does not encode the size of the noise

Use a mean-zero innovation with a regressor-dependent scale.



1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

$$Y = a + bX + \sigma(X)\varepsilon$$

$$E(u \mid X) = 0, \quad \mathrm{Var}(u \mid X) = \sigma(X)^2$$

Compare conventional and robust output

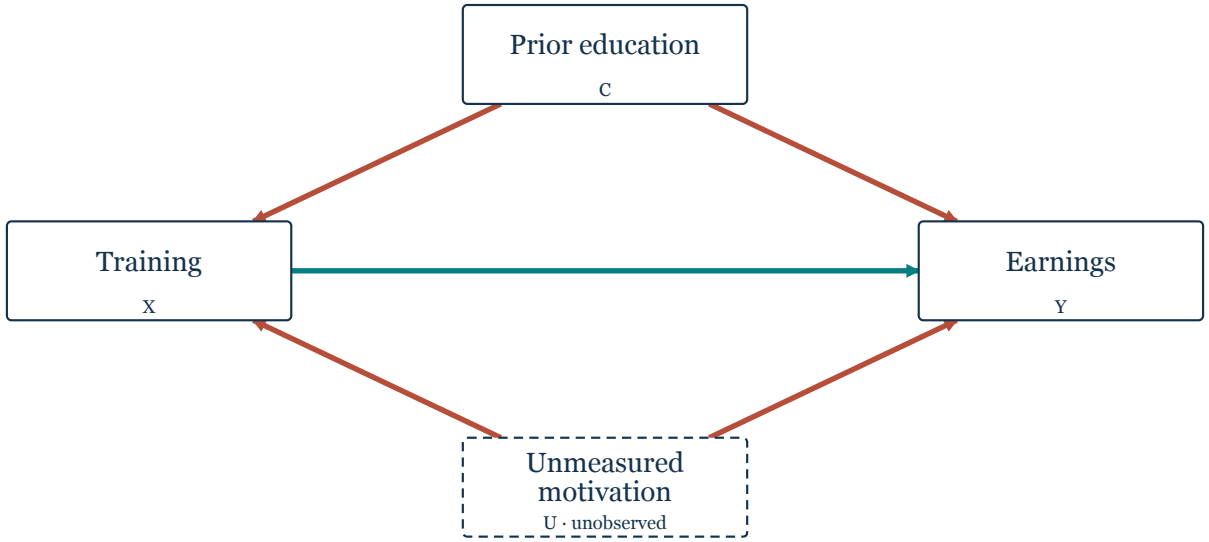
Same coefficients, different estimated uncertainty.

```
. regress score str el, vce(robust)

------------------------------------------------------------------------------
             |               Robust
       score | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         str |  -1.101296   .4328472    -2.54   0.011     -1.95213   -.2504616
          el |  -.6497768   .0310318   -20.94   0.000     -.710775   -.5887786
       _cons |   686.0322   8.728224    78.60   0.000     668.8754     703.189
------------------------------------------------------------------------------
```

Do not choose an SE by the resulting significance

Match it to the sampling and dependence structure.



1 open directed route · 2 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Robust does what?

Try the comparison before opening the explanation.

Practice

After changing regress y x to regress y x, vce(robust), which point estimate changes? Does this solve omitted-variable bias?

Robust does what?: explanation

Follow the calculation and interpret its units.

> On the same sample the OLS coefficient does not change. The estimated covariance matrix changes, so SEs, intervals and tests can change. Heteroskedasticity-robust inference does not remove omitted-variable bias or automatically allow serial dependence.

Test a substantively meaningful value

A zero null is not the only relevant question.

```
. test str=-1

 ( 1)  str = -1

       F(  1,   417) =    0.05
            Prob > F =    0.8151
```

Compute uncertainty for the policy contrast

Scale the SE along with the estimate.

```
. lincom -2*str

------------------------------------------------------------------------------
       score | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   2.202592   .8656943     2.54   0.011     .5009232     3.90426
------------------------------------------------------------------------------
```

$$\mathrm{SE}(-2\widehat{\beta}_1) = 2\,\mathrm{SE}(\widehat{\beta}_1)$$

A confidence interval can contain an important effect

Failing to reject is not proof of no effect.

> If an interval includes zero and also includes effects large enough to matter, the evidence may be too imprecise for the decision.

> Report the point estimate and interval, explain which practical values remain compatible with the model and discuss what data would improve precision.

Economic versus statistical size

Try the comparison before opening the explanation.

Practice

A precisely estimated policy effect is 0.001 outcome units. Is a small p-value enough to recommend the policy?

Economic versus statistical size: explanation

Follow the calculation and interpret its units.

> No. Report the units, realistic policy contrast, interval, policy cost and relevant outcome scale. Statistical detectability alone neither establishes causal validity nor a worthwhile benefit.

More observations help only under the relevant conditions

The information gain depends on what the observations add.

> Illustrative rate: $\text{SE} \propto n^{-1/2}$

Avoid repeated specification searches for a preferred p-value

The search changes what an isolated test can mean.

> If an analyst tries many outcomes, control sets and sample restrictions and reports only the most significant result, the reported p-value does not describe the whole search.

> Keep a clear account of the main question and intended specification.

Write an uncertainty statement for a policy reader

Include the comparison, units and conditions.

> A concise result sentence names the adjusted comparison, gives a feasible policy contrast and reports its uncertainty method.

> For example, describe a fitted two-student ratio difference at fixed language composition and give its robust interval in score points.

Check the reference distribution and scale

An interval is meaningful only for a defined quantity.

> Before interpreting an interval, identify whether it concerns a coefficient, a fitted contrast, an average outcome or an individual prediction.

> Also check whether the displayed interval uses a normal approximation or a t reference and whether the covariance option matches the observations.

Use the evidence to identify the next useful step

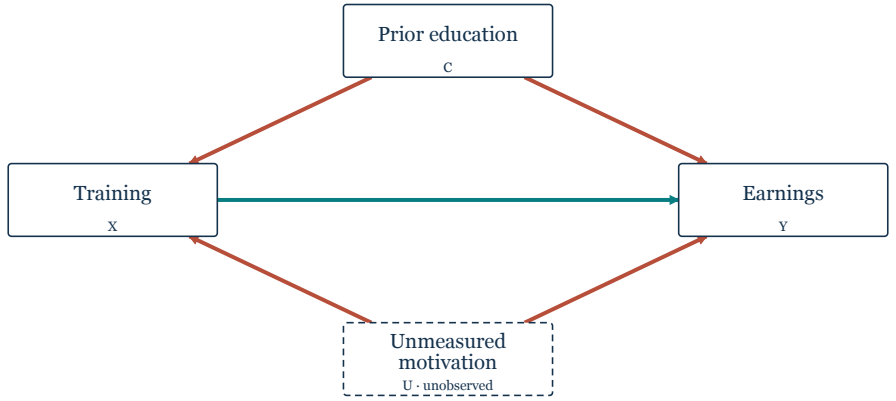
Precision, fit and design can require different improvements.

> An imprecise but credible comparison may benefit from more informative observations.

> These are different diagnoses.

DAG practice: uncertainty is not adjustment

Trace the routes before choosing a control.



1 open directed route · 2 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Select prior education as a control in the training diagram. Which alternative route remains? If the analyst changes conventional SEs to heteroskedasticity-robust SEs, does that close the route or change the training coefficient on an identical specification and sample?

DAG practice: uncertainty is not adjustment: explanation

Explain what conditioning changes in this stated graph.

> The route through unmeasured motivation remains open after conditioning on education. The common-cause arrows are properties of the proposed causal story, not of the covariance formula. Switching the SE option does not block them. On an unchanged model and sample, the OLS coefficient remains the same while the estimated covariance, intervals and tests can change. Robust inference addresses uncertainty under appropriate conditions. A credible causal comparison needs an additional argument about motivation or a design that addresses its role.

Report magnitude and uncertainty together

Reproduce the evidence, then explain its economic meaning.

Practice

Use the fitted two-student gain to assess a null gain of one score point, rather than zero. Calculate the corresponding test ratio and explain its economic meaning. Keep your commands, log, output and interpretation.

Report magnitude and uncertainty together: explanation

A reproducible result needs both a calculation and an interpretation.

> Two-student gain: 2.202592, robust SE 0.865694.

> Null gain of one: $(2.202592 - 1)/0.865694 = 1.389164$.

> Separate a policy-relevant magnitude from a test against zero.

Next, let the model represent percentages and curvature

The interpretation depends on how variables enter.

> We have connected a coefficient with its uncertainty and separated detectability from practical size.

> The next lesson changes the equation itself.